

WACO, TX, USA

WMO: 722560

Lat: 31.619N

Lon: 97.228W

Elev: 500

StdP: 14.43

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	27.9	11.7	10.2	32.1	15.7	12.4	37.0	27.4	50.9	24.9	53.1	8.6	0	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.3	101.4	74.1	99.3	74.7	97.3	74.8	78.4	91.0	77.8	90.3	77.2	89.4	10.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	136.2	81.8	74.8	132.8	81.5	74.1	129.5	81.3	42.3	91.3	41.5	90.0	40.7	89.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.0	20.8	19.0	18.6	104.7	3.9	3.2	15.7	106.9	13.4	108.8	11.2	110.6	8.4	112.9		
			16.8	80.1	3.6	1.7	14.2	81.3	12.1	82.3	10.0	83.2	7.4	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.6	48.1	52.0	58.9	66.7	74.8	82.2	85.7	85.9	79.3	68.9	58.1	49.5
	DBStd	15.54	9.81	10.33	9.18	7.55	6.33	4.38	3.56	4.06	6.19	8.20	9.38	9.49
	HDD50	434	153	90	29	2	0	0	0	0	0	1	32	127
	HDD65	2003	528	376	229	70	8	0	0	0	3	56	244	489
	CDD50	6855	95	147	306	502	770	966	1106	1111	878	588	275	111
	CDD65	2946	5	13	40	120	313	516	641	646	431	177	36	8
	CDH74	34281	34	96	266	947	2984	6151	8507	8595	4796	1637	228	40
	CDH80	17388	3	17	39	233	1171	3128	4819	4956	2393	590	36	3
Wind	WSAvg	9.6	9.7	10.2	10.8	11.1	10.4	9.9	9.5	8.6	7.7	8.7	9.2	9.1
Precipitation	PrecAvg	33.9	2.1	2.3	2.7	3.1	4.4	3.3	1.9	1.9	3.1	3.9	2.6	2.4
	PrecMax	59.5	6.1	7.9	9.8	9.4	15.0	12.0	8.6	10.3	9.5	15.2	9.7	9.8
	PrecMin	19.6	0.0	0.1	0.1	0.1	0.6	0.2	0.0	0.0	0.2	0.0	0.1	0.2
	PrecStd	8.0	1.5	1.5	1.8	2.2	3.2	2.6	2.1	2.2	2.3	3.4	1.9	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.4	82.1	84.3	90.3	96.1	101.9	103.4	104.6	101.5	93.3	84.3	78.8
		MCWB	60.8	62.1	65.9	69.5	73.5	74.5	74.1	73.5	72.0	71.6	68.1	63.9
	2%	DB	73.0	76.7	80.1	85.5	92.2	98.3	100.9	101.6	97.3	90.0	79.8	73.5
		MCWB	60.4	61.9	65.3	69.1	74.0	74.9	74.3	74.3	73.8	71.4	66.7	63.1
	5%	DB	68.9	72.3	76.9	82.3	89.4	95.5	98.8	99.6	94.4	86.5	76.3	69.6
		MCWB	59.0	61.2	64.4	68.4	73.3	75.2	74.9	74.8	73.4	69.7	64.6	60.9
	10%	DB	65.0	68.7	73.4	79.4	86.6	93.0	96.7	97.4	91.8	83.0	72.9	65.8
		MCWB	57.1	60.7	63.1	67.1	72.7	75.0	75.1	74.9	72.9	68.2	64.5	58.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.8	68.7	70.5	75.0	78.6	80.0	79.0	79.1	78.4	76.5	72.4	68.5
		MCDB	71.7	74.7	78.3	83.9	88.3	92.8	92.2	93.0	89.4	85.5	78.6	73.4
	2%	WB	64.5	66.4	68.7	72.8	76.9	78.5	78.1	78.1	77.3	74.9	70.1	66.0
		MCDB	69.1	71.5	75.5	81.3	86.3	90.6	91.5	91.8	88.0	83.2	75.3	70.7
	5%	WB	62.1	64.4	67.2	71.1	75.5	77.5	77.3	77.3	76.3	73.4	68.1	63.4
		MCDB	66.7	69.6	73.2	78.8	84.7	89.0	90.8	90.8	86.7	81.2	73.5	68.3
	10%	WB	58.4	61.8	65.4	69.6	74.3	76.7	76.7	76.7	75.4	72.1	65.8	59.9
		MCDB	63.6	67.6	71.3	76.4	83.4	87.6	89.6	89.6	85.5	79.2	71.8	64.6
Mean Daily Temperature Range	5% DB	MDBR	22.0	21.3	21.2	21.4	20.1	19.9	20.9	21.3	22.0	22.4	21.2	20.9
		MCDBR	28.1	26.1	23.5	22.9	21.8	22.3	23.3	23.7	23.9	24.6	23.6	25.2
		MCWBR	16.9	15.5	12.7	11.2	8.1	5.6	4.8	5.3	6.9	10.2	12.8	15.7
	5% WB	MCDBR	21.1	21.4	19.0	18.1	18.7	18.7	20.0	20.7	19.5	19.0	18.5	20.5
		MCWBR	15.7	15.4	12.4	10.6	8.4	6.3	5.3	5.5	6.5	9.3	13.3	15.4
Clear-Sky Solar Irradiance	taub	0.317	0.338	0.359	0.400	0.429	0.434	0.445	0.439	0.403	0.370	0.348	0.325	
	taud	2.505	2.453	2.410	2.317	2.296	2.318	2.322	2.323	2.371	2.447	2.456	2.487	
	Ebn at Noon	289	292	292	283	274	271	268	268	275	277	274	279	
	Edh at Noon	28	32	36	41	42	41	41	40	37	32	29	27	
All-Sky Solar Radiation	RadAvg	936	1084	1422	1722	1837	2084	2130	1971	1653	1318	987	823	
	RadStd	96	184	121	150	163	171	151	120	137	195	121	112	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only			N/A	N/A	N/A	N/A	N/A	-0.67	N/A	N/A	N/A
Regional (53 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page